DIGITAL TWIN-BASED INTELLIGENT TEMPERATURE CONTROL SYSTEM FOR PROCESS INDUSTRIES

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: INSTRUMENTATION AND CONTROL ENGINEERING

of Anna University

OCTOBER 2026

DEPARTMENT OF INSTRUMENTATION AND CONTROL ENGINEERING

PSG COLLEGE OF TECHNOLOGY

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SYNOPSIS

Temperature is an important process variable in thermal systems such as boilers, heaters and superheaters. Conventional temperature monitoring mainly provides the present process temperature, but it does not provide a synchronized virtual representation of the physical process or predict its future temperature behaviour.

This project proposes a Digital Twin-based intelligent temperature monitoring and prediction system for a kettle-based thermal process. An electronic kettle is used as the physical thermal process, while a DS18B20 digital temperature sensor measures the water temperature. An ESP32 microcontroller acquires the temperature data and performs real-time prediction using a mathematical temperature model and Linear Regression-based prediction methodology.

The developed system initially performs one-step-ahead prediction and has been extended to generate ten future temperature predictions from T+1 to T+10 using recursive prediction. An adaptive historical window is used, where actual temperature readings are used to update the prediction history. The predicted temperature values are compared with an experimental threshold to provide an early warning indication.

The present implementation achieves real-time temperature acquisition at approximately 1.03 s intervals. The obtained T+1 prediction error is low, with an MAE of 0.068 °C, while the error increases to 0.544 °C at T+10 due to recursive prediction. With the 80-reading prediction window, the error improves to 0.058 °C at T+1 and 0.395 °C at T+10. An experimental threshold of 85 °C provided approximately 12 seconds of early warning before the actual temperature crossed the threshold.

Industrial networks increasingly rely on the Message Queuing Telemetry Transport (MQTT) protocol, and malicious MQTT traffic can degrade the availability and integrity of industrial data. This project presents a software-based comparative study of three machine learning algorithms, Random Forest, Decision Tree and Isolation Forest, for detecting anomalous MQTT traffic in Industrial IoT communication.

A 3D model of the kettle has also been developed using SolidWorks and imported into Unity to form the virtual representation of the physical process. The integration of live

ESP32 temperature data with the Unity Digital Twin is currently in progress. Future work includes ML-based prediction, physical–virtual synchronization, cloud/IoT integration and closed-loop temperature control.

CONTENTS

CHAPTER **Page No.**

Acknowledgement ..... **i**
Synopsis ..... **ii**
List of Figures ..... **vi**
List of Tables ..... **vii**
1. INTRODUCTION 1
1.1 INTRODUCTION TO DIGITAL TWIN TECHNOLOGY 1
1.2 DIGITAL TWIN IN THERMAL PROCESS INDUSTRIES 2
1.3 REAL-TIME TEMPERATURE MONITORING AND PREDICTION 2
1.4 PROBLEM STATEMENT 3
1.5 NEED FOR THE STUDY 3
1.6 OBJECTIVES 3
1.7 SCOPE AND LIMITATIONS 4
1.8 LITERATURE SURVEY 4
1.9 RESEARCH GAP 5
2. SYSTEM DESCRIPTION AND DESIGN 6
2.1 SYSTEM OVERVIEW 6
2.2 SYSTEM ARCHITECTURE 6
2.3 ELECTRIC KETTLE AS A THERMAL PROTOTYPE 8
2.4 DESCRIPTION OF THE THERMAL PROCESS 9
2.5 ESP32-BASED DATA ACQUISITION AND PROCESSING UNIT 9
2.6 DS18B20 TEMPERATURE SENSOR AND MEASUREMENT PRINCIPLE 10
2.7 THERMAL ENERGY-BALANCE MODEL 10
2.8 PREDICTION METHODOLOGY 10
2.9 ADAPTIVE PREDICTION WINDOW 11
2.10 TEN-STEP FUTURE PREDICTION 12
3. IMPLEMENTATION AND METHODOLOGY 13
3.1 OVERALL METHODOLOGY 13
3.2 REAL-TIME DATA ACQUISITION 13
3.3 PREDICTION ERROR CALCULATION 14
3.4 THRESHOLD-BASED WARNING 15
3.5 SOLIDWORKS 3D MODEL DEVELOPMENT 15
3.6 UNITY INTERFACE DEVELOPMENT 16

4. RESULTS AND FUTURE WORK 17
4.1 REAL-TIME TEMPERATURE ACQUISITION AND PREDICTION 17
4.2 THRESHOLD-BASED EARLY WARNING 17
4.3 WEB INTERFACE AND CSV LOGGING 18
4.4 SOLIDWORKS AND UNITY-BASED DIGITAL TWIN 18
4.5 PHYSICAL-TO-VIRTUAL COMMUNICATION 19
4.6 MQTT-BASED COMMUNICATION 19
4.7 FUTURE INTELLIGENT CONTROL AND DIGITAL TWIN DEVELOPMENT 20
4.8 FUTURE WORK OF ON–OFF TEMPERATURE CONTROL 21
BIBLIOGRAPHY 22

LIST OF FIGURES

Fig 2.1 Block Diagram of the Proposed Real-Time Digital Twin for Temperature Monitoring and Prediction 7

Fig 2.2 ESP32 pin diagram..... 9

Fig 2.3 Actual temperature and ten-step-ahead predicted temperature. 12

Fig 3.1 Overall methodology of the proposed Digital Twin-based intelligent temperature monitoring and prediction system. 13

Fig 3.2 SolidWorks 3D model of the electric kettle imported into Unity 16

Fig 4.1 Variation of prediction error with forecast horizon. 17

Fig 4.2 Serial Monitor output showing ten-step temperature prediction and threshold status. 18

Fig 4.3 3D model of the electric kettle developed using SolidWorks and imported into Unity..... 19

Fig 4.4 MQTT subscriber receiving real-time temperature data from the ESP32. 20

Fig 4.5 MQTT publisher command used for temperature data transmission. 20

LIST OF TABLES

Table 1.1 Literature Survey 4
Table 2.1 Components of the System and Their Functions 7
Table 2.2 Implemented and Reported Parameters of the Thermal Prototype 8
Table 2.3 Reading number, Data used and purpose 11
Table 3.1 Temperature range and condition of process..... 15

CHAPTER 1
INTRODUCTION

Temperature is a critical variable in thermal processes such as boilers and superheaters. Conventional monitoring systems provide current temperature values but do not offer a continuously synchronized virtual representation capable of predicting future temperature behaviour. Digital Twin technology addresses this limitation by integrating real-time measurements with a mathematical model to monitor process behaviour, predict temperature changes, and identify potential deviations.

In this work, an electric kettle is used as a thermal prototype, with a DS18B20 temperature sensor and an ESP32 microcontroller for real-time data acquisition and processing. A thermal energy-balance model represents the temperature dynamics, while a Linear Regression model predicts future temperature values using historical measurements. The system supports real-time monitoring, ten-step temperature prediction, threshold-warning demonstration, and data logging. A three-dimensional kettle model has also been developed in SolidWorks and imported into Unity, with live data synchronization and closed-loop temperature control identified as future developments.

1.1 INTRODUCTION TO DIGITAL TWIN TECHNOLOGY

A Digital Twin is a virtual representation of a physical system that uses real-time data and mathematical models to monitor and analyse the behaviour of the actual process. It combines sensor measurements, data processing, and prediction techniques to understand system performance and estimate future operating conditions. In thermal process industries, Digital Twin technology can support temperature monitoring, predictive analysis, and early identification of potential deviations, helping improve process understanding and operational efficiency.

In this project, an electric kettle is used as a thermal prototype to demonstrate real-time temperature monitoring and prediction. An ESP32 microcontroller acquires temperature data from a DS18B20 sensor, while a thermal energy-balance model represents the heating behaviour and a Linear Regression model predicts future temperature values. The system generates one-step and ten-step predictions, calculates prediction errors, records temperature data, and demonstrates threshold warnings. A three-dimensional kettle model has also been developed using SolidWorks and imported into Unity to support the visualization component of the Digital Twin.

1.2 DIGITAL TWIN IN THERMAL PROCESS INDUSTRIES

Thermal process industries involve operations in which heat transfer and temperature regulation play an important role in maintaining process efficiency and product quality. Examples include boilers, heat exchangers, furnaces, power plants, and chemical processing systems. Digital Twin technology can be applied to these processes by combining real-time sensor measurements with mathematical models and predictive algorithms. This approach helps monitor temperature variations, analyse heating behaviour, and estimate future process conditions. It can also support early warnings when the measured or predicted temperature approaches a predefined threshold.

In a thermal Digital Twin, the physical process provides temperature data through sensors, while the virtual model uses the collected data to represent and predict the process behaviour. In this project, an electric kettle is used as a simplified thermal prototype to demonstrate these concepts. The ESP32 acquires temperature readings from the DS18B20 sensor, and the collected data is processed using a thermal energy-balance model and Linear Regression-based prediction. The predicted temperature values are compared with actual measurements to evaluate accuracy and demonstrate early threshold warnings.

1.3 REAL-TIME TEMPERATURE MONITORING AND PREDICTION

Real-time temperature monitoring is an important aspect of thermal process management, as it provides continuous information about the temperature changes occurring in a physical system. By collecting sensor readings at regular intervals, the heating behaviour of the process can be observed and analysed. Temperature prediction extends this capability by estimating future temperature values from previously recorded data. Combining real-time monitoring with predictive analysis helps identify temperature trends, evaluate prediction accuracy, and generate early warnings when the predicted temperature approaches a predefined threshold.

In this project, real-time temperature monitoring is implemented using a DS18B20 temperature sensor connected to an ESP32 microcontroller. Temperature readings are acquired at approximately one-second intervals and processed to generate one-step-ahead and ten-step-ahead predictions using a Linear Regression model. The predicted values are compared with actual temperature measurements, and the Mean Absolute Error (MAE) is calculated to evaluate prediction performance. A threshold-warning mechanism is also implemented to indicate when the predicted temperature approaches the selected threshold of 85°C. These features demonstrate how real-time data

acquisition and temperature forecasting can be integrated to improve the monitoring and analysis of a thermal process.

1.4 PROBLEM STATEMENT

Conventional temperature monitoring systems mainly display the current temperature but may not predict future temperature changes. This makes it difficult to identify potential threshold violations in advance. Integrating real-time monitoring with predictive analysis can help overcome this limitation.

This project develops a Digital Twin of an electric kettle using an ESP32 and DS18B20 temperature sensor. A thermal energy-balance model and Linear Regression are used to predict future temperatures, calculate prediction errors, and generate threshold warnings. The system aims to demonstrate the feasibility of real-time temperature monitoring and prediction using a virtual representation of a thermal process.

1.5 NEED FOR THE STUDY

Real-time temperature monitoring alone may not provide sufficient information about the future behaviour of a thermal process. Predicting temperature changes in advance can help identify potential threshold violations and improve process monitoring. Therefore, integrating sensor data, mathematical modelling, and predictive techniques is useful for developing an effective Digital Twin system.

This project uses an electric kettle as a thermal prototype to demonstrate real-time temperature acquisition and prediction. By combining an ESP32, DS18B20 sensor, thermal energy-balance model, and Linear Regression, the system evaluates temperature behaviour and generates early threshold warnings. The study provides a foundation for understanding how Digital Twin technology can be applied to industrial thermal processes.

1.6 OBJECTIVES

The main objective of this project is to develop a Digital Twin for real-time temperature monitoring and prediction using an electric kettle as a thermal prototype. The specific objectives are:

- To acquire real-time temperature data using an ESP32 and DS18B20 sensor.
- To develop a mathematical model based on the thermal energy-balance equation.
- To predict future temperature values using Linear Regression.
- To evaluate prediction accuracy using Mean Absolute Error (MAE).

- To implement threshold-warning functionality for early temperature alerts.
- To develop a 3D kettle model using SolidWorks and support Digital Twin visualization through Unity.

1.7 SCOPE AND LIMITATIONS

The scope of this project includes real-time temperature monitoring, mathematical modelling, future temperature prediction, and threshold-warning generation using an electric kettle as a thermal prototype. The system uses an ESP32 and DS18B20 sensor for data acquisition, a thermal energy-balance model to represent heating behaviour, and Linear Regression to predict temperatures up to ten steps ahead. Prediction accuracy is evaluated using Mean Absolute Error (MAE), while SolidWorks and Unity support the development of the 3D visualization component.

The system is limited to the selected thermal prototype and the available temperature dataset. Prediction accuracy may vary with changes in heating conditions, sensor measurements, and forecast horizon. The selected 85°C threshold is an experimental value used to demonstrate warning generation, not an industrial safety limit. Full real-time synchronization of the 3D model and closed-loop temperature control remain areas for further development.

1.8 LITERATURE SURVEY

The review presentation identifies Digital Twin frameworks, thermal prediction and heat-transfer fundamentals as the conceptual background for this project. The literature is used to motivate the physical–virtual relationship and the need for a defensible thermal model.

Table 1.1 Literature Survey

Source	Relevance recorded in project materials
F. Tao, H. Zhang, A. Liu and A. Y. C. Nee, "Digital Twin in Industry: State-of-the-Art," IEEE Transactions on Industrial Informatics, 2019.	Digital Twin concepts and industrial applications.
R. Ala-Laurinaho et al., "Data Link for the Creation of Digital Twins," IEEE Access, vol. 8, pp. 228675–228684, 2020.	Data links supporting Digital Twin creation.
Y. A. Çengel and A. J. Ghajar, Heat Transfer: Fundamentals and Applications, 5th ed., McGraw-Hill, 2015.	Heat-transfer fundamentals used to frame thermal energy balance.

Source	Relevance recorded in project materials
F. Laamarti et al., "An ISO/IEEE 11073 Standardized Digital Twin Framework for Health and Well-Being in Smart Cities," IEEE Access, vol. 8, pp. 105950–105961, 2020.	Example of a standardized Digital Twin framework.

The presentation also lists a paper titled "Digital Twin for Temperature Field Prediction in a Boiler System" by Sanghyun Park, Jonghoon Lee and Kwangho Kim in Applied Thermal Engineering (2021). Full bibliographic details were not supplied in the source files; they should be verified before formal submission.

1.9 RESEARCH GAP

From the reviewed literature, Digital Twin technology has been widely studied for industrial monitoring and smart manufacturing. However, there is scope for combining:

- Real-time thermal sensing,
- Future temperature prediction,
- 3D virtual process representation,
- Threshold-based early warning, and
- Future ML-based intelligent prediction

within a single thermal-process Digital Twin prototype.

The present project addresses this by developing a kettle-based physical system and its corresponding virtual representation, with real-time temperature prediction as the primary functionality.

CHAPTER 2
SYSTEM DESCRIPTION AND DESIGN

The proposed system develops a Digital Twin for real-time temperature monitoring and prediction using an electric kettle as a thermal prototype. It combines sensor data acquisition, mathematical modelling, and machine learning to analyse temperature behaviour. The system also generates future temperature predictions and threshold warnings. A 3D model supports the virtual representation of the thermal process.

2.1 SYSTEM OVERVIEW

The proposed system integrates real-time temperature monitoring, mathematical modelling, and predictive analysis to develop a Digital Twin of a thermal process. An electric kettle is used as the physical prototype, with an ESP32 microcontroller and DS18B20 temperature sensor for temperature data acquisition. The collected readings are processed to analyse the heating behaviour and estimate future temperature values.

The system uses a thermal energy-balance model to represent the heating process and a Linear Regression model to generate one-step and ten-step temperature predictions. Prediction accuracy is evaluated using Mean Absolute Error (MAE), and a threshold-warning mechanism provides early alerts when the predicted temperature approaches 85°C. Additionally, a three-dimensional kettle model developed using SolidWorks and imported into Unity supports the virtual representation of the physical system. These components form the foundation of the proposed Digital Twin framework.

2.2 SYSTEM ARCHITECTURE

The system architecture consists of four major stages: temperature data acquisition, data processing, temperature prediction, and virtual visualization. The DS18B20 sensor measures the temperature of the water in the electric kettle and sends the readings to the ESP32 microcontroller. The collected temperature data is processed using a thermal energy-balance model and a Linear Regression algorithm to predict future temperature values.

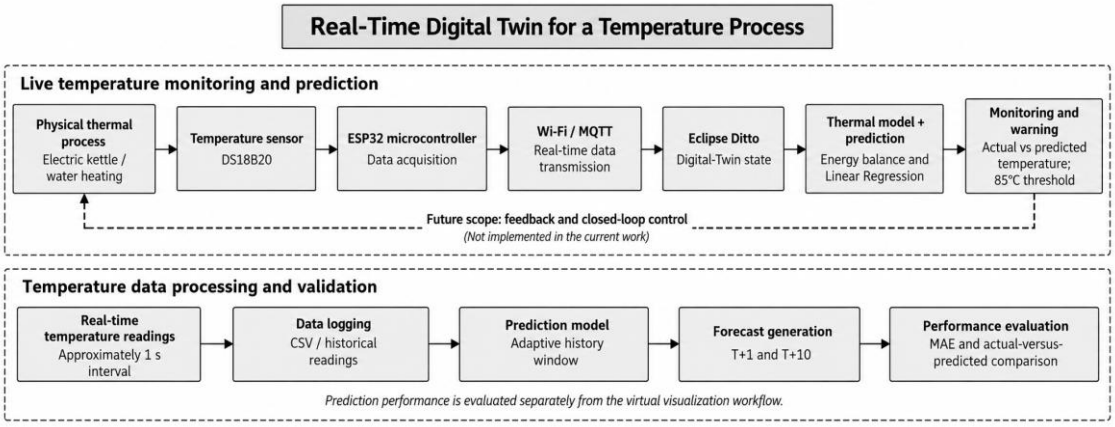


Fig 2.1 Block Diagram of the Proposed Real-Time Digital Twin for Temperature Monitoring and Prediction

The predicted temperature values are compared with actual measurements to evaluate the system performance using Mean Absolute Error (MAE). A threshold-warning mechanism generates alerts when the predicted temperature approaches the predefined limit of 85°C. The temperature data and prediction results are also used to support the virtual representation of the kettle through a 3D model developed using SolidWorks and imported into Unity. This architecture integrates the physical thermal process with mathematical modelling and predictive analysis to form the proposed Digital Twin system.

Table 2.1 Components of the System and Their Functions

Component	Function	Status
Electric Kettle	Acts as the physical thermal process for water heating	Implemented
DS18B20 Temperature Sensor	Measures real-time water temperature	Implemented
ESP32 Microcontroller	Acquires sensor readings and performs real-time processing and prediction	Implemented
Thermal Energy-Balance Model	Represents the heating behaviour of the thermal process	Implemented
Linear Regression Model	Predicts future temperature using historical actual readings	Implemented
Adaptive Historical Window	Selects the available readings or rolling 10/80-reading window for prediction	Implemented

Component	Function	Status
CSV Data Logging	Stores temperature readings and prediction data for analysis	Implemented
Serial Monitor / Web Interface	Displays actual temperature, predictions, and warning status	Implemented
Threshold-Warning Mechanism	Generates warnings based on the experimental 85°C threshold	Implemented
SolidWorks	Creates the 3D model of the electric kettle	Completed
Unity	Imports and displays the 3D kettle model; live-data synchronization is in progress	Physical – Virtual : Completed Virtual – Physical : In progress
Cloud / MQTT Integration	Supports future real-time communication with the Digital Twin platform	Future Work
Closed-Loop Control	Uses feedback to adjust the thermal process based on predicted temperature	Future Work

2.3 ELECTRIC KETTLE AS A THERMAL PROTOTYPE

The prototype is a Prestige PKOSS 1.5 electric kettle with a stated heater power of 1350 W and a water quantity of approximately 1.5 L (approximately 1.5 kg). The reported ambient/reference temperature is 30.31 °C. This setup supports observation of a heating transient in a controlled, small-scale system.

Table 2.2 Implemented and Reported Parameters of the Thermal Prototype

Parameter	Implemented / reported value
Thermal prototype	Prestige PKOSS 1.5 electric kettle
Heater power	1350 W
Water quantity	1.5 L (approximately 1.5 kg)
Temperature sensor	DS18B20
Controller	ESP32 DevKit
Sensor interface / pin	1-Wire / GPIO 4
Ambient/reference temperature	30.31 °C
Sampling interval	Approximately 1 s
Live experimental readings	527

2.4 DESCRIPTION OF THE THERMAL PROCESS

The thermal process involves heating water inside an electric kettle by converting electrical energy into thermal energy. The temperature of the water increases over time as heat is transferred from the heating element to the water. The rate of temperature rise depends on the heater power, quantity of water, specific heat capacity, and heat lost to the surrounding environment.

In this project, the DS18B20 temperature sensor measures the water temperature, and the ESP32 collects the readings at approximately one-second intervals. The measured data is used to analyse the heating behaviour and develop a thermal energy-balance model. Linear Regression is then applied to predict future temperature values using historical readings. This integration of real-time measurements, mathematical modelling, and temperature prediction supports the development of the proposed Digital Twin system.

2.5 ESP32-BASED DATA ACQUISITION AND PROCESSING UNIT

The ESP32 DevKit serves as the primary data acquisition and processing unit in the proposed system. It interfaces with the DS18B20 temperature sensor through the 1-Wire communication protocol using GPIO 4. The controller acquires temperature readings at approximately one-second intervals and updates the historical data required for real-time temperature prediction.

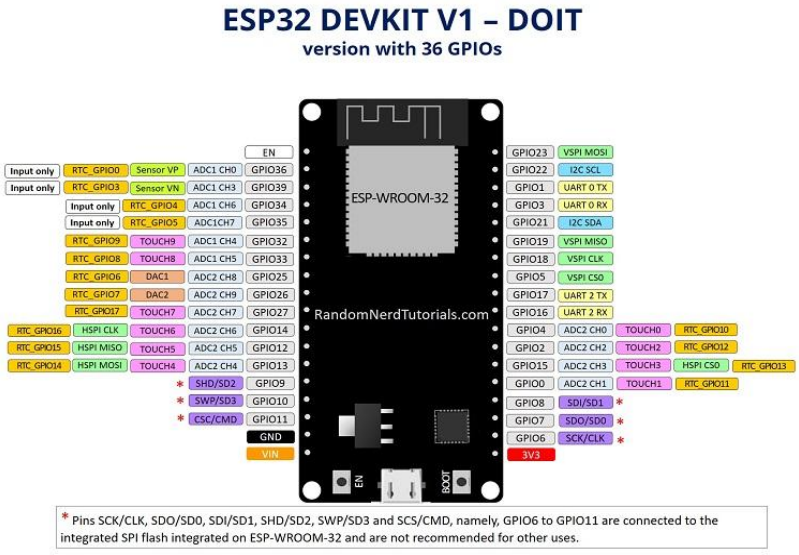


Fig 2.2 ESP32 pin diagram

The ESP32 processes the acquired temperature readings using the implemented Linear Regression model to generate one-step and ten-step future predictions. It also supports prediction-error calculation, threshold-warning generation, and CSV data logging. The

measured temperature and predicted values are displayed through the Serial Monitor and web interface for continuous monitoring of the thermal process.

2.6 DS18B20 TEMPERATURE SENSOR AND MEASUREMENT PRINCIPLE

The DS18B20 is a digital temperature sensor used to measure the water temperature in the electric kettle. It communicates with the ESP32 through the 1-Wire protocol, using GPIO 4 for data transmission. The sensor provides digital temperature readings, which are acquired at approximately one-second intervals for real-time monitoring and prediction.

The measured temperature values are processed by the ESP32 and used as historical inputs for the Linear Regression model. These readings support one-step and ten-step temperature prediction, prediction-error calculation, and threshold-warning generation. The DS18B20 therefore acts as the primary measurement element connecting the physical thermal process with the computational prediction system.

2.7 THERMAL ENERGY-BALANCE MODEL

The thermal energy-balance model is used to represent the temperature variation of water inside the electric kettle during the heating process. The model considers the electrical heating power supplied to the water, the thermal energy required to increase its temperature, and the heat lost to the surrounding environment. The relationship is expressed using the following equation:

$$P = mC_p \frac{dT}{dt} + UA(T - T_a) \tag{2.1}$$

where P is the heating power, m is the mass of water, C_p is the specific heat capacity of water, $\frac{dT}{dt}$represents the rate of temperature change, UA represents the overall heat-loss coefficient, T is the water temperature, and T_a is the ambient temperature.

For implementation in the digital twin, the continuous thermal model is represented in discrete time so that it can be used with the temperature readings collected at approximately one-second intervals. The measured temperature data are processed to study the heating behaviour and support temperature prediction. This mathematical representation provides a basis for understanding the physical thermal process and comparing the model-based behaviour with the actual sensor measurements.

2.8 PREDICTION METHODOLOGY

The temperature prediction system uses historical temperature measurements obtained from the DS18B20 sensor to estimate the future temperature of the thermal process. The

basic idea is that the recent temperature behaviour contains information about the near-future behaviour of the process. Linear Regression is used to establish a mathematical relationship between the previous temperature measurements and the next temperature value. The prediction equation is given by

$$\hat{T}(t+1) = \beta_0 + \sum_{i=0}^{p-1} \beta_{i+1} T(t-p+1+i) \quad (2.2)$$

where $\hat{T}(t+1)$ represents the predicted next temperature, T represents the actual temperature measurements, p is the number of historical readings used for prediction, β_0 is the regression intercept and β_i represents the regression coefficients.

The temperature readings are arranged in chronological order and a historical window is selected as the input to the prediction model. For every new sensor reading, the available actual temperature history is used to predict the next temperature value. The model initially performs one-step-ahead prediction, represented as $T+1$. The actual DS18B20 readings are continuously used to update the historical data, while the predicted values are kept separate from the actual-history buffer. This allows the prediction model to remain based on real sensor measurements and provides a reliable basis for subsequent future-temperature prediction.

2.9 ADAPTIVE PREDICTION WINDOW

The prediction accuracy of a time-series model depends on the amount of historical information provided to the model. In the developed system, an adaptive historical window strategy is used instead of maintaining a single fixed window throughout the experiment. The size of the prediction window is selected according to the current reading number. During the initial stage, when only a limited number of measurements are available, all the available actual temperature readings are used to gradually build the required history.

Table 2.3 Reading number, Data used and purpose

Reading Number	Historical Data Used	Purpose
1–9	All available actual readings	Build initial history
10–79	Latest 10 actual readings	Short-term rolling prediction
80 onwards	Latest 80 actual readings	Long-term rolling history

From readings 10–79, the latest 10 actual temperature measurements are used to represent the recent temperature trend of the thermal process. From reading 80 onwards, the latest 80 actual temperature measurements are used as the historical window, providing a larger amount of temperature information for the prediction model.

This allows the system to adapt the amount of historical data used for prediction as more measurements become available.

2.10 TEN-STEP FUTURE PREDICTION

The prediction system is extended from one-step prediction to generate ten future temperature values from **T+1 to T+10**. For every new actual sensor reading, the appropriate historical window is selected and the Linear Regression model first predicts $T+1$. The predicted $T+1$ value is then used internally to calculate $T+2$, and this recursive process continues until $T+10$ is obtained.

This is a recursive prediction process in which each future prediction provides the input required for calculating the following prediction. After generating all ten predicted values, the system identifies the maximum predicted temperature and compares it with the selected temperature threshold. This allows the system to determine whether the temperature is expected to approach or exceed the threshold within the next ten prediction steps.

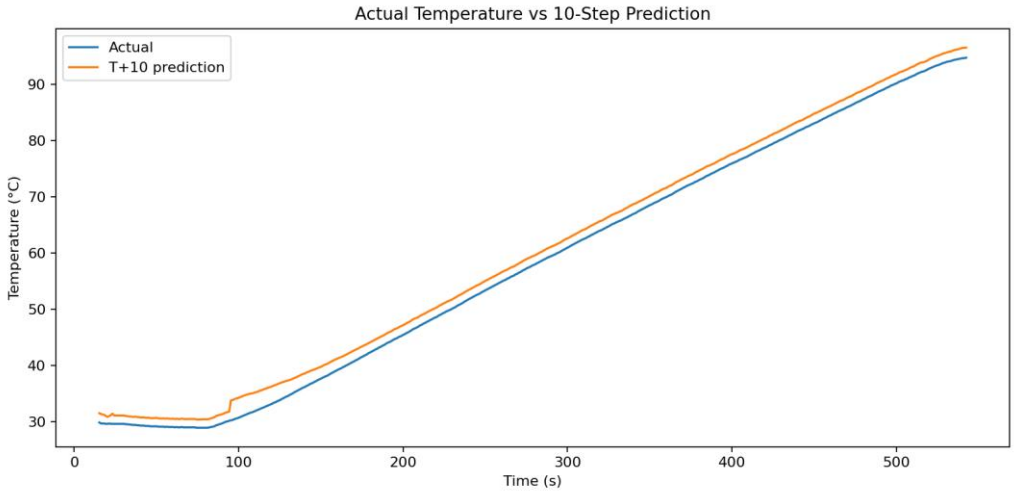


Fig 2.3 Actual temperature and ten-step-ahead predicted temperature.

The ten-step prediction provides a short-term view of the future temperature behaviour and supports early identification of an increasing temperature trend. However, the prediction error generally increases as the prediction horizon increases because errors in earlier predictions can propagate into subsequent predictions. The developed system therefore uses the ten-step forecast primarily for predictive monitoring and early warning of approaching temperature conditions.

CHAPTER 3
IMPLEMENTATION AND METHODOLOGY

3.1 OVERALL METHODOLOGY

The proposed system integrates real-time temperature measurement, data acquisition, temperature prediction and Digital Twin visualization to develop an intelligent monitoring system for a thermal process. An electronic kettle is used as the physical thermal process, while a DS18B20 digital temperature sensor measures the water temperature. The measured temperature is acquired by the ESP32 and processed in real time. The collected temperature values are maintained as historical data and used as input for the Linear Regression-based prediction model.

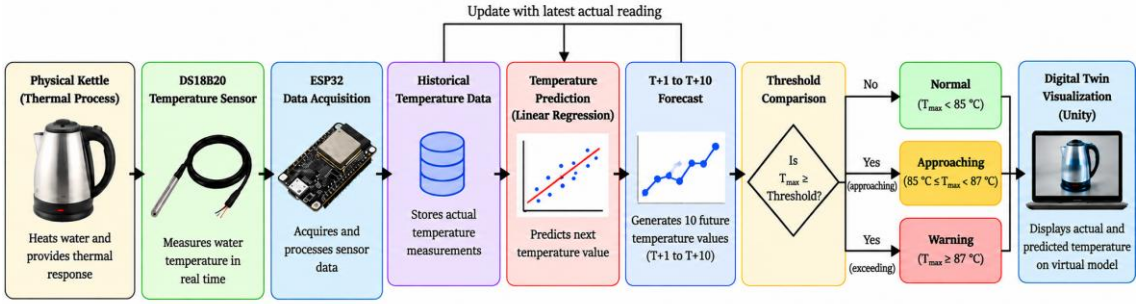


Fig 3.1 Overall methodology of the proposed Digital Twin-based intelligent temperature monitoring and prediction system.

For each new sensor reading, the system predicts the future temperature from T+1 to T+10 using the selected historical temperature window. The maximum predicted temperature is then compared with the predefined threshold to determine the process condition and provide an early indication of an increasing temperature trend. The measured and predicted temperature data are intended to be transferred to the Unity-based virtual model for Digital Twin visualization. The current real-time implementation operates at approximately a 1-second sampling interval and provides continuous temperature prediction and monitoring through the ESP32-based system.

3.2 REAL-TIME DATA ACQUISITION

The DS18B20 digital temperature sensor is used to measure the temperature of the water inside the electronic kettle. The sensor provides a digital temperature measurement, which is acquired by the ESP32 microcontroller. The ESP32 acts as the main data-acquisition unit and processes the temperature information received from the sensor. The sensing and computation cycle operates at approximately **1.03 seconds**

per reading, allowing the temperature variation of the thermal process to be monitored in near real time.

Each temperature measurement is associated with its corresponding reading sequence and stored chronologically. The collected measurements form the historical temperature dataset required for prediction. As new measurements are obtained, the latest actual temperature value is added to the historical data while older values are retained according to the selected prediction window. This continuously updated dataset provides the input required for the Linear Regression prediction model.

The real-time data-acquisition process therefore establishes the connection between the physical thermal process and the computational prediction system. The sequence can be represented as **Kettle → DS18B20 → ESP32 → Temperature Data → Historical Data**, providing the basic data stream required for subsequent prediction and Digital Twin development.

3.3 PREDICTION ERROR CALCULATION

The performance of the temperature prediction model is evaluated by comparing the predicted temperature with the corresponding actual temperature measured by the DS18B20 sensor. Since the prediction system generates multiple future temperature values, the prediction error is evaluated separately for different forecast horizons, ranging from T+1 to T+10. This makes it possible to determine how prediction accuracy changes as the prediction horizon increases.

The Mean Absolute Error (MAE**)** is used as one of the main evaluation metrics. It represents the average absolute difference between the actual and predicted temperature values and is calculated using:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|T_i - \hat{T}_i\right| \tag{3.1}$$

where T_i represents the actual temperature$\hat{T}_i$ represents the predicted temperature, and n represents the number of evaluated samples. A lower MAE indicates better prediction accuracy.

The calculated error values are used to compare the performance of one-step and multi-step predictions. In general, the prediction error can increase with the forecast horizon because later predictions depend on previously predicted values in the recursive forecasting process. Therefore, evaluating T+1 through T+10 provides a better understanding of the reliability of the prediction model for short-term temperature forecasting.

3.4 THRESHOLD-BASED WARNING

A threshold-based warning mechanism is incorporated to demonstrate how the prediction system can be used for early identification of an increasing temperature condition. Instead of considering only the current measured temperature, the system examines the maximum temperature predicted within the next ten forecast steps. The predicted maximum temperature is then compared with the selected experimental threshold.

For the current demonstration, an experimental threshold of **85 °C** is considered. The process condition is classified based on the predicted maximum temperature as follows:

Table 3.1 Temperature range and condition of process

Predicted Maximum Temperature	Process Condition
$T_{max} < 85°C$	Normal
$85°C \leq T_{max} < 87°C$	Approaching
$T_{max} \geq 87°C$	Warning

This approach provides an early indication before the actual temperature reaches the selected threshold. If the predicted temperature indicates that the process is approaching or exceeding the threshold, an appropriate warning condition can be generated.

3.5 SOLIDWORKS 3D MODEL DEVELOPMENT

A three-dimensional model of the physical electronic kettle was developed using SolidWorks to create the virtual representation required for the Digital Twin system. The model represents the major physical features of the kettle, including the body, lid, handle and other visible structural elements. The purpose of developing the model is to create a virtual counterpart that can represent the physical thermal process in a computer-based environment.

After completing the 3D modelling process, the SolidWorks model was converted into a format suitable for use in the Unity development environment. The model was then imported into Unity 6, where it forms the basic visual structure of the virtual kettle. This enables the physical kettle and its corresponding virtual representation to be considered as two connected parts of the proposed Digital Twin system.

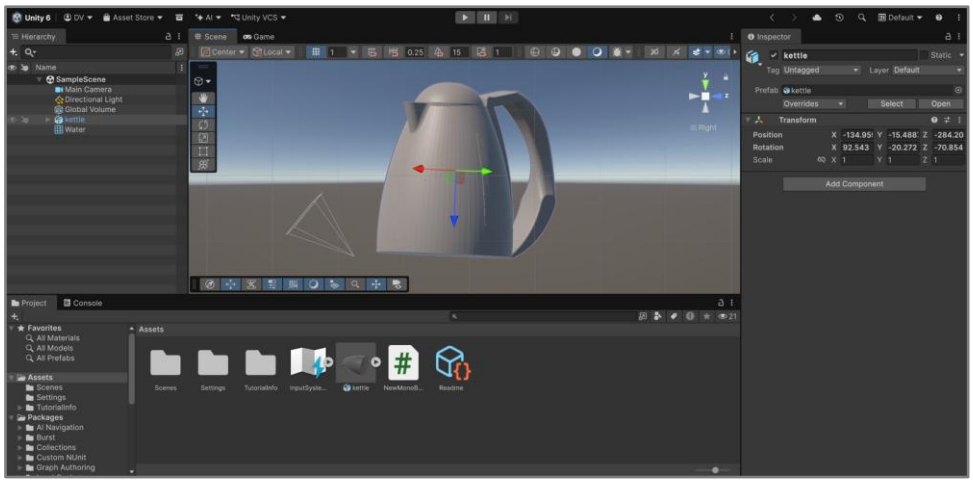


Fig 3.2 SolidWorks 3D model of the electric kettle imported into Unity

The developed 3D model provides the foundation for future integration of real-time temperature information. Once the communication link is established, the virtual model can be updated using the temperature information obtained from the physical kettle, allowing the Digital Twin environment to represent the current state of the thermal process.

3.6 UNITY INTERFACE DEVELOPMENT

The completed SolidWorks kettle model was imported into Unity 6 to develop the virtual environment for the Digital Twin. The Unity scene contains the virtual kettle and water objects, providing a visual representation of the physical thermal process. The model was positioned and configured in Unity to provide the foundation for real-time visualization of the physical kettle.

The implemented data flow is:

Physical Kettle → DS18B20 → ESP32 → Communication Link → Unity → Virtual Kettle / Temperature Display

At the current stage, communication is implemented only in the kettle-to-Unity direction. Therefore, real-time process information can be transferred from the physical system to the virtual environment, while control commands or data from Unity back to the physical kettle have not yet been implemented.

CHAPTER 4
RESULTS AND FUTURE WORK

4.1 REAL-TIME TEMPERATURE ACQUISITION AND PREDICTION

The developed system successfully acquires real-time temperature measurements from the DS18B20 sensor through the ESP32-based data-acquisition system. The sensing and computation cycle operates at approximately 1.03 seconds per reading, allowing the temperature variation of the kettle to be continuously monitored. The collected actual temperature measurements are stored chronologically and used as input for the prediction system.

The prediction results demonstrated that the system can provide short-term temperature forecasting. The obtained T+1 MAE was 0.068 °C, while the T+10 MAE was 0.544 °C. When the larger 80-reading historical window was used, the prediction accuracy improved, with T+1 MAE of 0.058 °C and T+10 MAE of 0.395 °C. As the forecast horizon increases, the error generally increases because the multi-step prediction is performed recursively. Future work will focus on evaluating additional ML models and improving long-horizon prediction accuracy using a larger temperature dataset.

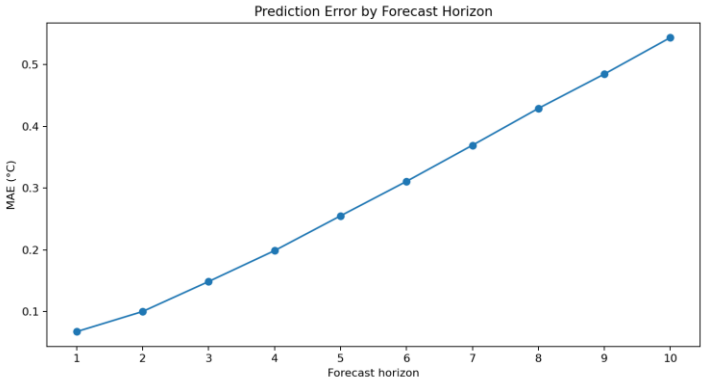


Fig 4.1 Variation of prediction error with forecast horizon.

4.2 THRESHOLD-BASED EARLY WARNING

The ten-step prediction was further used to demonstrate an early-warning mechanism. The maximum predicted temperature among T+1 to T+10 was compared with the experimental threshold. During the experimental demonstration, the system generated the first warning at approximately reading 437, while the actual temperature crossed 85 °C at reading 449, providing an early indication of approximately 12 seconds.

The current threshold is used only to demonstrate the prediction-based warning functionality and is not considered an industrial safety limit. In future work, the threshold

can be determined based on the actual operating requirements and safety constraints of the thermal process. The warning mechanism can also be extended to generate different levels of alarms and eventually initiate automatic control actions.

4.3 WEB INTERFACE AND CSV LOGGING

The ESP32 web page displays the latest actual temperature, selected history-window size, ten future predictions, maximum predicted temperature, threshold and status. The interface also provides access to CSV data. The Serial Monitor presents corresponding runtime information.

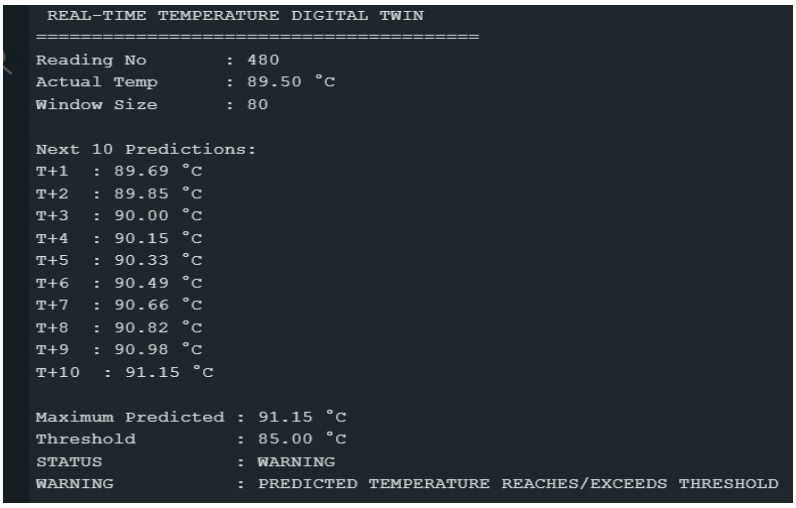


Fig 4.2 Serial Monitor output showing ten-step temperature prediction and threshold status.

The source documentation includes an interface snapshot at a reading where the actual temperature was 89.50 °C, the window size was 80 and the maximum predicted temperature was 91.15 °C against an 85.00 °C threshold, resulting in WARNING status. This is an example runtime snapshot, not a summary of the entire experiment.

4.4 SOLIDWORKS AND UNITY-BASED DIGITAL TWIN

A three-dimensional model of the physical kettle was successfully developed using **SolidWorks** and imported into **Unity 6**. The Unity environment contains the virtual kettle and water objects, providing a visual representation of the physical thermal process. This establishes the virtual model required for the Digital Twin implementation.

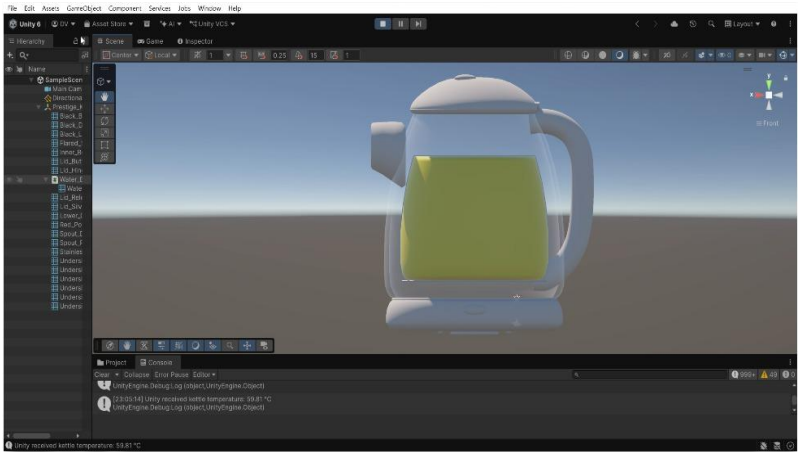


Fig 4.3 3D model of the electric kettle developed using SolidWorks and imported into Unity.

The next stage is to further develop the Unity environment so that it can display additional process information such as actual temperature, predicted temperature and process status. The virtual model can also be enhanced with animations and graphical indicators to represent changes in the physical process more clearly.

4.5 PHYSICAL-TO-VIRTUAL COMMUNICATION

A one-way communication link from the physical kettle to Unity has been successfully implemented. The DS18B20 measures the kettle temperature, the ESP32 acquires the sensor data, and the temperature information is transmitted to the Unity application.

This establishes the initial physical-to-virtual connection required for the Digital Twin. At the present stage, communication is one-way only; Unity does not yet send commands back to the physical kettle. Future work will focus on completing bidirectional communication, where Unity can transmit control commands to the ESP32 and the physical process can respond accordingly.

4.6 MQTT-BASED COMMUNICATION

MQTT communication was implemented using the Mosquitto broker to establish a reliable data-transfer pathway between the ESP32-based physical system and the computer. The ESP32 publishes the measured temperature using the topic kettle/temperature, and the temperature values were successfully received by the computer through an MQTT subscriber.

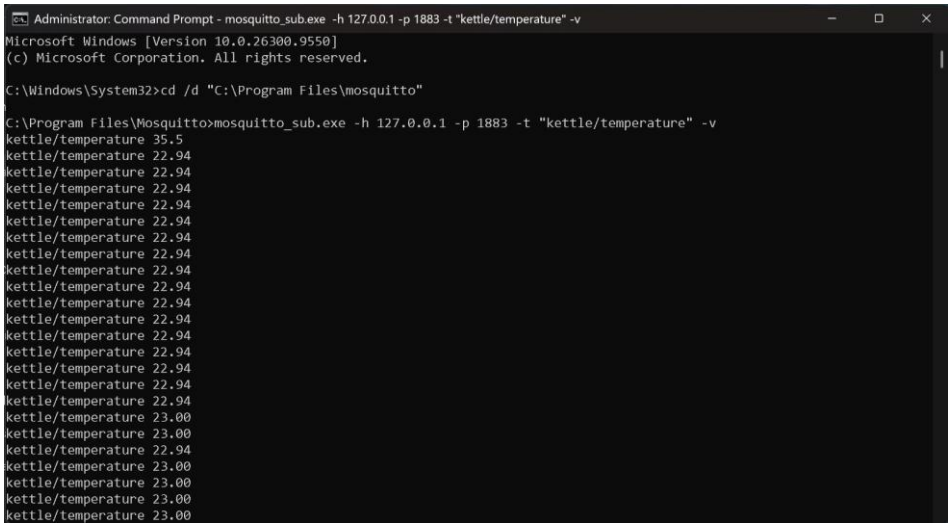


Fig 4.4 MQTT subscriber receiving real-time temperature data from the ESP32.

The successful MQTT communication provides a foundation for continuous physical–virtual synchronization. Future development will integrate the MQTT data directly into the Unity application so that the virtual kettle automatically updates according to the real-time physical temperature. This can subsequently be extended to transmit predicted temperature values and process-status information.

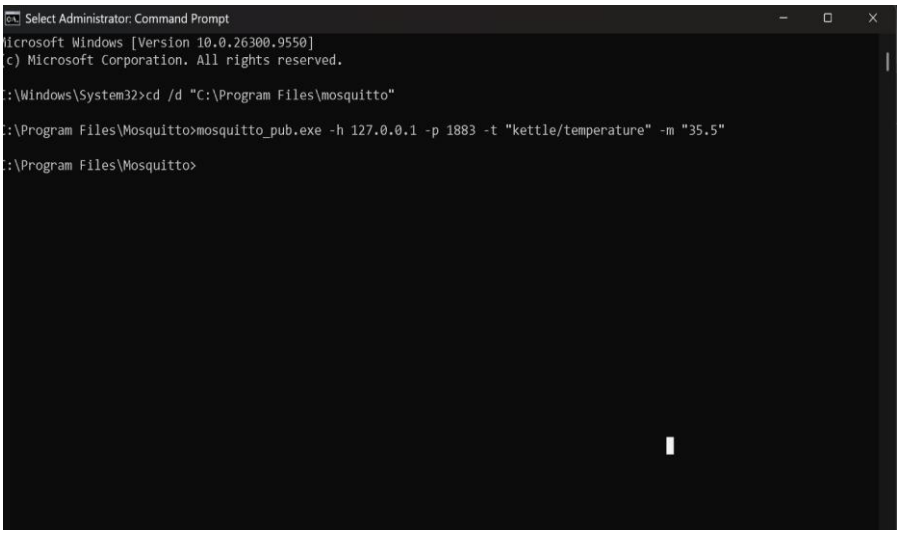


Fig 4.5 MQTT publisher command used for temperature data transmission.

4.7 FUTURE INTELLIGENT CONTROL AND DIGITAL TWIN DEVELOPMENT

The current system mainly performs real-time sensing, prediction, warning and physical-to-virtual data transfer. The next stage is to integrate the collected temperature dataset with more advanced Machine Learning techniques and evaluate their performance using metrics such as MAE, RMSE and R^2. The prediction model can then be integrated into the real-time Digital Twin environment.

Future development will also focus on bidirectional communication and closed-loop temperature control. In the proposed future system, the Digital Twin can receive real-time measurements from the physical kettle, predict its future temperature, and generate an appropriate control command. This command can then be sent back to the ESP32 and used to control the heating process. Thus, the system can be extended from a monitoring and prediction platform into an intelligent closed-loop Digital Twin-based temperature-control system.

4.8 FUTURE WORK OF ON–OFF TEMPERATURE CONTROL

The next stage of the project is to implement an ON–OFF temperature controller for the physical kettle. The controller will use the measured temperature from the DS18B20 sensor and compare it with a predefined temperature setpoint. Based on the temperature error, the controller will generate a control action to switch the heating element ON or OFF. When the temperature is below the desired setpoint, the heater will be switched ON, and when the temperature reaches or exceeds the specified limit, the heater will be switched OFF.

The ON–OFF control strategy will provide the first closed-loop control implementation for the developed thermal system. The control action generated by the ESP32 can be used to operate the heating element through an appropriate switching/driver circuit. The temperature response can then be monitored through the Unity Digital Twin, allowing the physical and virtual systems to be observed together. In a later stage, the ON–OFF controller can be enhanced using hysteresis to reduce frequent switching of the heater and improve temperature stability.

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